Title + Problem Statement & Solution

Real Estate CRM System

Data Engineering Capstone Project

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Problem Statement:

Real estate businesses manage large volumes of data across agents, clients, properties, and transactions. Without a structured data system, tracking leads, measuring agent performance, and analyzing market trends becomes inefficient.

Solution:

A complete Real Estate CRM system with a normalized PostgreSQL database, automated ETL pipeline, analytics queries, and an interactive dashboard for business insights.

TECH STACK & ARCHITECTURE

Tech Stack:

🛠️ Tech Stack

Layer	Technology
Database	PostgreSQL 15 (Docker)
Data Generation	Python, Faker
ETL Pipeline	Python, Pandas, psycopg2
Analytics	SQL (Window functions, JOINs, Aggregations)
Dashboard	Streamlit
Backend API	FastAPI, Uvicorn
Containerization	Docker, Docker Compose

🏗️ Architecture & Data Flow(diagram):

```
┌──────────────────────────────────────────────────────────────┐
│                          DATA SOURCES                        │
│          CSV Files            +        Faker (Seed)          │
└──────────────────────────────────────────────────────────────┘
              │                              │
              ▼                              ▼
┌──────────────────────┐      ┌──────────────────────────┐
│   ETL Pipeline       │      │       Seed Script        │
│  etl/etl_pipeline.py │      │      etl/seed.py         │
│                      │      │                          │
│  Extract (CSV)       │      │  Generate fake data      │
│  Transform (pandas)  │      │  Insert via psycopg2     │
│  Load (psycopg2)     │      │                          │
└──────────────────────┘      └──────────────────────────┘
              │                              │
              └──────────────┬───────────────┘
                             ▼
┌──────────────────────────────────────────────────────────────┐
│                     PostgreSQL Database                      │
│                     (Docker Container)                       │
│                                                              │
│  agents │ clients │ properties │ listings │ leads           │
│                 transactions │ interactions                  │
└──────────────────────────────────────────────────────────────┘
                             │
              ┌──────────────┴──────────────┐
              ▼                             ▼
┌──────────────────────┐      ┌──────────────────────────┐
│  FastAPI             │      │  Streamlit Dashboard     │
│  backend/            │      │  dashboard/app.py        │
│  main.py             │      │                          │
│                      │      │  KPI Cards               │
│  GET /agents         │      │  Agent Rankings          │
│  GET /props          │      │  Price by City           │
│  GET /trans          │      │  Lead Conversion         │
└──────────────────────┘      └──────────────────────────┘
```

SCREENSHOTS(part 1)

7 tables :

```
              List of relations
 Schema |     Name      | Type  | Owner
--------+---------------+-------+-------
 public | agents        | table | admin
 public | clients       | table | admin
 public | interactions  | table | admin
 public | leads         | table | admin
 public | listings      | table | admin
 public | properties    | table | admin
 public | transactions  | table | admin
(7 rows)
```

ETL pipeline output:

```
Clients before cleaning: 10
Properties before cleaning: 10
Clients after cleaning: 9
Properties after cleaning: 8
Clients loaded : 9
Properties loaded : 8
ETL pipeline complete!
```

Seed script output:

```
Agents inserted: 20
Clients inserted: 109
Properties inserted: 73
Listings inserted : 60
Leads inserted :100
Transaction inserted : 40
Interactions inserted : 120
```

Total revenue per agent:

full_name	total_revenue
Gregory Chambers	38550285.92
Donald Johns	13069279.64
Erik Zavala	12127704.81
Jonathan Fernandez	10188680.74
Lisa Johnson	9936907.52

(5 rows)

Payment mode:

payment_mode	total
cash	15
cheque	13
loan	12

(3 rows)

Top agents by listings:

full_name	total_listings
Debra Smith	8
Erik Zavala	6
Andrea Ibarra	6
Gregory Chambers	5
Donald Johns	4

(5 rows)

Avg price by city:

city	avg_price
West William	9839738.50
East Craig	9825206.39
North Dannyfort	9377466.85
Lake George	9201294.31
Dustinburgh	9016869.74

(5 rows)

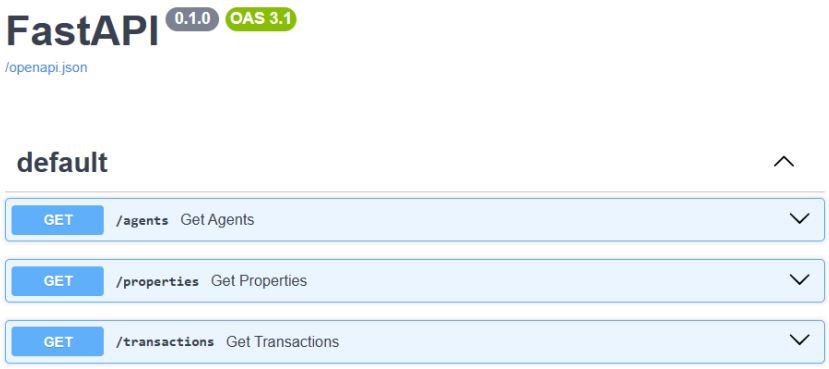
Lead conversion:

title	total_leads	total_transactions
Rise challenge health.	6	3
Spring interesting yourself.	5	0
Beyond spring form friend.	4	4
Must place.	4	0
Believe strong lay final.	4	10

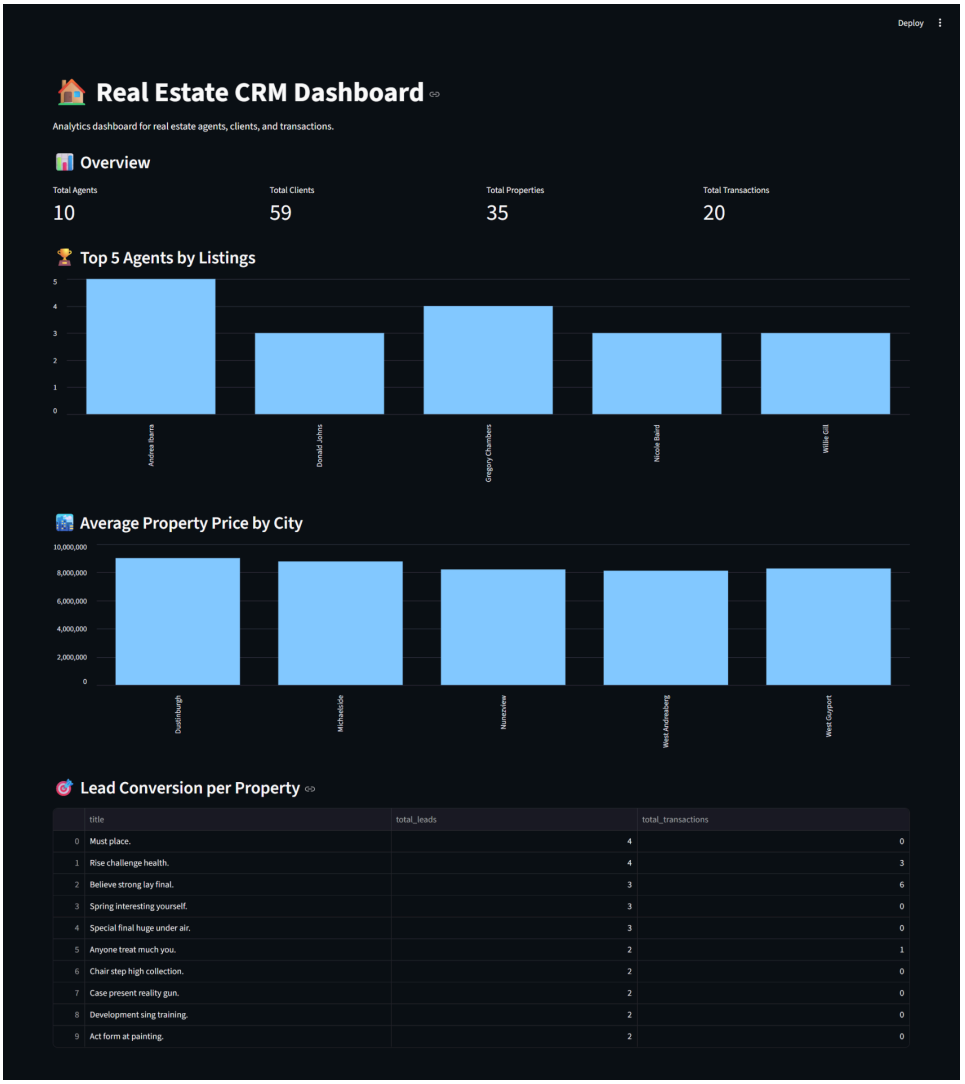
(5 rows)

SCREENSHOTS(part 2)

Fastapi routes:



Streamlit dashboard:



Unique Points & Future Improvements:

Unique Points:

1. End-to-end data engineering pipeline from raw CSV ingestion to live analytics dashboard, demonstrating the complete data lifecycle.
2. Dirty data simulation and cleaning in the ETL pipeline — invalid enums, missing fields, and negative values are detected and dropped before loading.
3. Dockerized PostgreSQL ensures the entire database layer is portable and reproducible — any reviewer can run the project with a single docker compose up command.
4. Seven normalized tables with foreign key constraints and CHECK constraints enforce data integrity at the database level.
5. REST API layer built with FastAPI exposes the database over HTTP with auto-generated Swagger documentation.

Future Improvements:

1. Real-time data ingestion using Apache Kafka to replace the batch CSV pipeline with a streaming architecture.
2. Authentication and role-based access control for FastAPI endpoints using JWT tokens.
3. Cloud deployment on AWS RDS for PostgreSQL and EC2 for the application layer.
4. Replace Streamlit dashboard with a Next.js frontend for a more production-grade user interface.
5. Add Apache Airflow DAGs to schedule and monitor the ETL pipeline automatically.